

XUANZHE HAN

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EDUCATION

University of California, Los Angeles (UCLA)

Master of Science in Computer Science

Bachelor of Science in Computer Science

Sep 2024 – Jun 2026 (Expected)

Sep 2020 – Jun 2024

SKILLS

Languages: C/C++, Python, Java, TypeScript/JavaScript, SQL

ML: PyTorch, scikit-learn

Systems/Infra: CUDA, ROS, Docker, Git, Linux

Web/Backend: React, Flask, FastAPI; Postgres, MongoDB

EXPERIENCE

Nissan North America, Inc.

Systems Research Intern

Sep 2024 – Mar 2025

- Built a configurable ROS + C++ pipeline to align sensor data and generate training-ready features for context-aware routing (pedestrian density, speed, obstructions, school zones, etc.).
- Contributed to an internal dashboard by building dataset indexing and query logic to support visualization of collected AV data (FastAPI, TypeScript/JavaScript).
- Collaborated with researchers to define data schemas and add new derived signals/features as needs evolved.

Tap Series, LLC

Web Development Intern

Oct 2023 – Jan 2024

- Built a React + Flask analytics dashboard to parse sales and customer-interaction data and visualize trends by time, region, and demographics.
- Added interactive filtering and drill-down views to support internal reporting and analysis workflows.
- Integrated Twilio SendGrid API to ingest email event analytics and surface campaign performance metrics in the dashboard.

Huazhong University of Science and Technology

Research Intern, Facial Action Unit Detection

Aug 2023 – Sep 2023

- Adapted JAANet baselines to new facial expression datasets to enable standardized comparisons for ongoing research.
- Built and benchmarked PyTorch and scikit-learn pipelines to evaluate action unit detection performance.
- Authored experiment documentation and literature review summaries to support publication preparation.

RELATED COURSEWORK

Foundations: Algorithms & Complexity, Design of Programming Languages

Systems & Networks: Operating Systems, Computer Architecture, Compilers, Parallel & Distributed Computing, Network Architecture & Protocols

AI & Data: Computer Vision, Natural Language Processing, Reinforcement Learning, On-Device LLM Deployment (llama.cpp), ML Optimization, Probabilistic Models, Data Management Systems

EXTRACURRICULAR ACTIVITIES

3D4E (3D Printing Club)

Webmaster & Team Lead

2022 – 2024

UCLA

- Maintained the club website (3d4eat UCLA.github.io); shipped content/UX updates and kept documentation current to support member onboarding.
- Led biomedical and mechanical keyboard subteams; designed and delivered functional prototypes (e.g., pulse oximeter, custom macro pad), coordinating requirements, fabrication, and iteration.

PROJECTS

DecentraDocs

JavaScript, React, WebRTC, Yjs

- Implemented a peer-to-peer collaborative document editor with real-time multi-user sync using Yjs CRDTs over a custom PeerJS/WebRTC transport.
- Integrated authenticated user identities via Dex OIDC and documented practical connectivity constraints (NAT/firewalls requiring rendezvous/relay infrastructure).

Ownly

Named Data Networking, TypeScript/JavaScript, Rust, Go, OpenMLS

- Ongoing: secure decentralized workspace built on the Named Data Networking (NDN) stack (UCLA, Prof. Lixia Zhang).
- Implementing an “MLS-lite” group messaging scheme with OpenMLS to simplify key usage by encrypting directly with the evolving group secret.

P-Forest

Python, scikit-learn

- Implemented a network intrusion detection system based on *p-forest: In-Network Inference with Random Forests* and evaluated on CIC-IDS-2017.
- Generated TCAM/CRAM rules plus RMT simulator inputs; used benchmarking tooling to compare performance.

Dynamic Kafka

Python, Docker, Kafka

- Prototyped dynamic batching in Apache Kafka by containerizing brokers, producers, and consumers with Docker Compose.
- Built Python producers with adjustable batching and automated topic-management scripts; explored auto-tuning to study latency-throughput tradeoffs.

On-Device LLM Deployment

llama.cpp, Quantization, LoRA, ADB, Perffetto

- Deployed and benchmarked small LLMs on an edge device using llama.cpp; built an on-device evaluation workflow (ADB automation, throughput + power profiling) to compare fine-tuning and quantization tradeoffs.

CUDA RNN

C++, CUDA

- Implemented an RNN from scratch and accelerated training with custom CUDA kernels (matrix ops + activations).
- Optimized memory layout and GPU execution to improve training throughput versus a CPU-only baseline.

SlangLingo

React, OpenAI API

- Built a slang-aware translation web app powered by a fine-tuned GPT-3.5 model to handle informal and idiomatic user inputs.

ResuMatch

Chrome Extension, JavaScript

- Built a Chrome extension that extracts LinkedIn job requirements and matches them against uploaded resumes to streamline applications.